

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[]args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("Enter the number of elements:");
6         int n=sc.nextInt();
7         int[] arr=new int[n];
8         System.out.println("Enter the elements in sorted order:");
9         for(int i=0;i<n;i++){
10            arr[i]=sc.nextInt();
11        }
12        System.out.print("Enter the element to search:");
13        int key=sc.nextInt();
14        int low=0,high=n-1;
15        boolean found=false;
16        while(low<=high){
17            int mid=(low+high)/2;
18            if(arr[mid]==key){
19                System.out.println("Element found at index: "+ mid);
20                found=true;
21                break;
22            }else if(arr[mid]<key){
23                low=mid+1;
24            }else{
25                high=mid-1;
26            }
27        }
28        if(!found){
29            System.out.println("Element not found.");
30        }
31        sc.close();
32    }
33 }

```

```

Enter the number of elements:Enter the elements in sorted order:
Enter the element to search:Element found at index: 2

```